

MD. RAKIB HOSSAIN

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PROFESSIONAL SUMMARY

Final-year Computer Science and Engineering student working across data and software. I turn messy, real-world data into clear insights — cleaning it, analysing it, building machine-learning models, and reporting the results — and I build and ship complete applications across web, desktop, and embedded systems. Highlights: a live machine-learning web app I deployed, a 2,300-image dataset I built and annotated from scratch, and a robotics team I founded that has won national and international awards. Seeking a role in data analytics, data science, or software development.

TECHNICAL SKILLS

Data and Analytics: Python, SQL, pandas, NumPy, Microsoft Excel, Power BI

Machine Learning: scikit-learn, PyTorch, TensorFlow, CatBoost, XGBoost, YOLO, OpenCV, SHAP

Software Development: C++, JavaScript; Next.js, NestJS, .NET, REST APIs, PostgreSQL, SQL Server

Embedded: Arduino C++, ESP32, sensor interfacing

Tools: Git, GitHub, VS Code, Jupyter, Roboflow; AI-assisted development with Claude Code and Antigravity

EDUCATION

Bachelor of Science in Computer Science and Engineering

Expected 2026

American International University-Bangladesh (AIUB), Dhaka

Higher Secondary Certificate (HSC), Dhaka Imperial College, 2021

Secondary School Certificate (SSC), Dhaka Collegiate School, 2019

Languages: Bangla (Native), English (Fluent)

PROJECTS

RideGuard — Data Analysis and Real-Time ML Web Application

Python, SQL, CatBoost, XGBoost, TensorFlow, SHAP

- Cleaned and engineered **two raw datasets** — resolving missing values, noise, and inconsistent formats — and geocoded locations to build an analysis-ready pipeline.
- Performed exploratory analysis to identify the behavioural and spatial factors driving risk, then engineered features from those findings.
- Trained and calibrated gradient-boosting models (CatBoost, XGBoost) against deep-learning baselines, tuning hyperparameters and evaluating with appropriate metrics; applied SHAP so each prediction is explainable.
- Built and **deployed the full application** (TypeScript frontend, Python model service) with a live risk map, route-level scoring, and voice and vibration alerts.

Live: <https://ride-guard-web-app-web.vercel.app> | Code: https://github.com/rakib4123/Ride_Guard_WebApp

DhakaNight — Dataset Creation and Model Benchmarking

Python, PyTorch, YOLO, OpenCV, Roboflow

- Built and annotated a custom dataset of **approximately 2,300 low-light street images** in Roboflow, addressing a gap where existing datasets performed poorly at night.
- Trained a YOLO detection model and ran a controlled benchmark of four image-enhancement methods (CLAHE, Gamma, Zero-DCE, RetinexFormer), quantifying each method's impact on detection accuracy.

Code: https://github.com/rakib4123/Dhaka_Night

Drikon — Full-Stack E-Commerce Platform

Next.js, NestJS, PostgreSQL, Prisma, Redis

- Architected and built a production-quality platform as a monorepo: Next.js storefront, NestJS REST API, PostgreSQL with Prisma, and Redis caching.
- Implemented secure authentication — two-factor login, rotating tokens, and role-based access control — plus a separate admin panel for product and order management.
- Deployed entirely on free-tier cloud services at **zero monthly cost**.

Code: <https://github.com/rakib4123/drikon>

AIUB STEAM — University Portal (Desktop Application)

C#, .NET, WinForms, SQL Server

- Developed a Windows desktop portal with role-based dashboards for students, faculty, and administrators, backed by a SQL Server database.
- Implemented course pre-registration, assignment uploads, and grade viewing, with email-OTP sign-up and an admin console for account approval.

Code: <https://github.com/rakib4123/AiubSteam>

PulseStone — Embedded Device and Statistical User Study

ESP32-C3, Arduino C++

- Designed and built a handheld ESP32-C3 device from the ground up, writing firmware in Arduino C++ that reads sensor input and drives synchronized LED, haptic, and on-screen feedback in real time.

- Designed and ran a controlled within-subjects user study (15 participants, 150 trials) and analyzed the results with a paired t-test, demonstrating a **14.5x faster start time** than a smartphone app ($p < 0.001$).

Code: <https://github.com/rakib4123/PulseStone>

LEADERSHIP AND ACTIVITIES

Founder and Team Lead — “Khepa Chakka” Robotics Team

2023 — Present

- Founded and lead a competitive robotics team, taking robots from concept to the competition floor.
- Designed, built, and programmed multiple classes of robot — line-following, soccer, race, and battle bots — covering sensor integration, motor control, and real-time embedded logic.
- Coordinate team members across mechanical design, electronics, and software, dividing tasks and managing timelines to meet competition deadlines.
- Led the team to **multiple national and international podium finishes**, including selection for Techfest, IIT Bombay.

Active Member — Engineering Students' Association of Bangladesh (ESAB), AIUB

2023 — Present

- Collaborate with faculty on research and development; participate in seminars and workshops on emerging technologies.

AWARDS AND HONORS

- **Champion** — Robot Soccer, AIUB CS Fest 2024.
- **Winner** — Cozmo Clench segment, Techfest, IIT Bombay (Bangladesh Zonal) 2024, one of Asia's largest science and technology festivals.
- **1st Runner-up** (Nano Battle Bot) and **2nd Runner-up** (Robo Race) — National Robotics Championship 2025.
- **1st Runner-up** — Robo Race, Technoxian Bangladesh National Round 2024.
- **1st Runner-up** — Robot Soccer, AIUB Robotic Crew (ARC).
- Competed in numerous additional national and inter-university robotics competitions, including BUET Autofest 2024 (Soccer Bot and Robo Race).